Week 10: Payment Rails and Dollar Dominance

FINA3020 at CUHK Business School

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Lecture Overview

[Background Context]

- Payment rails, ledgers, and the difference between clearing and settlement
- Domestic fast payment systems and interoperability
- Why cross-border payments are slow, costly, and risky
- Correspondent banking, prefunding cost, and settlement risk
- Corporate treasury, payment providers, and investment-account funding
- Dollar dominance, external balance sheets, and joint asset/FX probability models

The Microstructure of Payments

[Core Concept]

A payment rail is the infrastructure through which a debtor pays a creditor. Common rails:

- Cash, bank transfers, and wire transfers
- Card networks such as Visa, Mastercard, and American Express
- Instant payment systems and mobile wallets

Most non-cash rails distinguish several stages:

- **Authorization**: verifying that the payer may initiate the payment
- **Clearing**: transmitting, validating, and sometimes netting payment obligations
- **Settlement**: discharging those obligations by transferring the settlement asset

Settlement finality is a legal as well as operational concept. A settled bank transfer may be irrevocable, whereas a card payment can remain subject to refunds or chargebacks even after interbank settlement.

Who Keeps the Ledger?

[Core Concept]

Every electronic rail must maintain a consistent record of balances and prevent unauthorized or duplicate payments.

- A **ledger** is a record of transaction history used to validate new payments.

Centralized ledger

- A trusted third party keeps the record.
- Examples: commercial banks, a central bank, a card network.
- Can be fast and scalable, but creates governance, operational-resilience, and trust requirements.

Distributed ledger

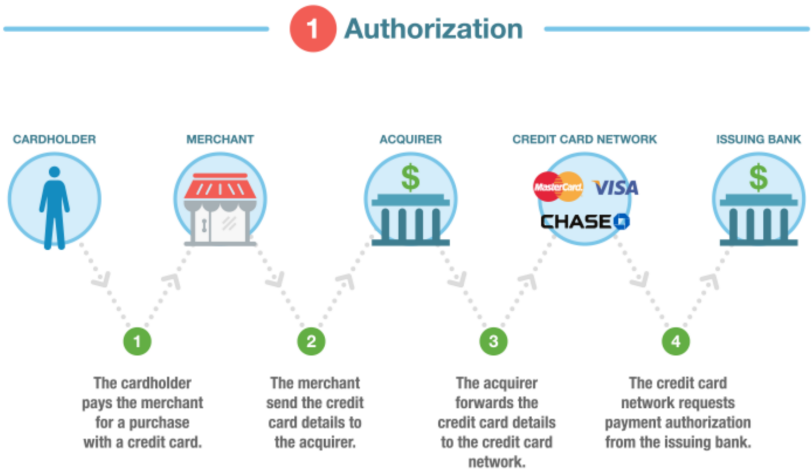
- Participants share the record and agree on it through a protocol.
- Examples include permissioned ledgers and public blockchains using proof of work or proof of stake.
- Trades reliance on one operator for protocol, validator, governance, and interoperability risks. Performance and resource use depend on the design.

Ledger Economics Trilemma →

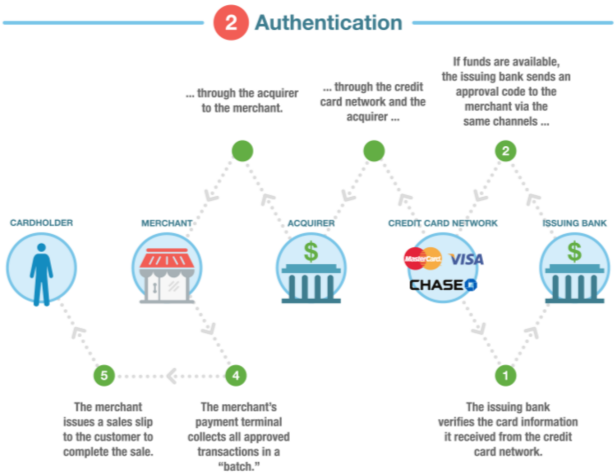
Card Payments: Authorization through Settlement [Core Concept]

- Authentication checks the identity or credentials involved in a transaction. Authorization asks the issuing bank whether to approve the payment.
- The merchant sends the transaction through its acquirer and card network to the issuer. An authorization can reserve purchasing capacity without transferring final funds.
- Clearing reconciles transaction data and calculates obligations. Settlement discharges those obligations through transfers between institutions.
- The merchant's payout timing and chargeback exposure depend on its contract. Approval at the terminal is not the same as final, irrevocable settlement.

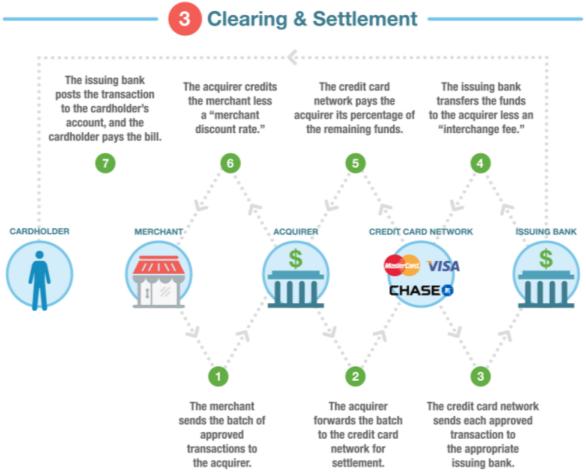
Card Networks Illustration: Authorization



Card Networks Illustration: Authentication



Card Networks Illustration: Settlement



Wires, ACH, and Instant Payment Systems [Core Concept]

- Fedwire transfers settle individually in central-bank money with legal finality under its rules. CHIPS uses liquidity-saving arrangements for large-value USD payments.
- ACH handles batch credit and debit payments, including payroll and bills. Settlement can be same day or on later business days, depending on service and cutoff.
- FedNow and the private RTP network provide instant-payment infrastructure for participating US institutions. A payment app is a customer interface, not necessarily the underlying settlement rail.
- Speed, operating hours, finality, error recovery, and consumer protection are separate dimensions. A fast payment may still be fraudulent or sent to the wrong person.

US payment-system institutions →

Domestic Fast Payment Systems

[Background Context]

Many jurisdictions now have instant payment systems operated by central banks, public utilities, or industry consortia.

- Brazil PIX, Hong Kong FPS, India UPI, and US FedNow

Shared design features:

- **Always on**: customer transfers are available 24/7. Interbank settlement arrangements differ
- **Rapid settlement**: the payee usually receives usable funds in seconds, subject to scheme and legal rules
- **Broad reach**: participating banks and payment providers can transfer across a common scheme
- **Convenient addressing**: many systems support aliases or QR codes, although designs differ
- **Potentially low cost**: pricing to users and merchants depends on regulation and provider strategy

Policy goals: reduce cash use, cut transaction costs for small firms, and widen financial inclusion.

Country Systems in Detail →

PIX, FPS, and UPI: Country Designs

[Background Context]

- Brazil's PIX launched in 2020 with central-bank infrastructure and common participation and addressing arrangements. Broad participation helped establish network reach.
- Hong Kong's FPS launched in 2018, connecting participating banks and stored-value facilities for HKD and RMB transfers, with convenient aliases and QR interfaces.
- India's UPI connects participating bank accounts through interoperable applications under NPCI and RBI oversight. App competition sits on shared payment infrastructure.
- Fees, access rules, and protections differ by provider and scheme. Historical adoption statistics describe a dated snapshot, not permanent pricing or universal coverage.

PIX, FPS, and UPI details → FPS and UPI →

Instant Payments in Brazil: PIX

[Background Context]



Interoperability and Financial Inclusion

- Systems such as PIX, FPS, and UPI let people pay from existing bank or wallet accounts, using supported devices and common interfaces.
- A common acceptance standard can let merchants receive from customers at many participating institutions, which can raise competition.
- Some systems link aliases or identity infrastructure to regulated accounts. Financial inclusion still depends on access, onboarding, and consumer protection.

Interoperability increases the size and reach of the network. It can also reduce the incentive for a private firm to invest in a proprietary payment network.

Why Cross-Border Payments Are Harder

[Core Concept]

A domestic rail settles inside one currency and one legal system. A cross-border payment loses that.

- **Variable speed and finality**: compliance, cut-off times, time zones, and multiple ledgers can delay a payment, although some corridors now complete in seconds.
- **Currency risk**: the exchange rate can move between when the payment starts and when it settles.
- **Layered costs**: providers may charge fees or embed margins in the exchange rate. Routing and fee incidence vary by corridor.
- **Legal and compliance risk**: public authorities can block payments, while private institutions may delay or reject them to meet sanctions and anti-money-laundering obligations.

These payments settle the trade and capital flows from Week 6, so their cost affects real activity.

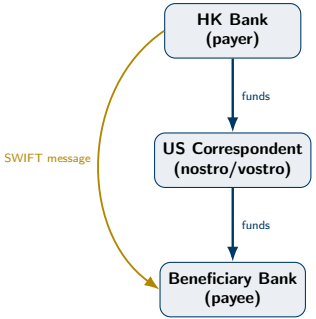
Correspondent Banking Rails

[Core Concept]

The traditional rail chains banks together.

- **SWIFT** sends the payment message between banks. It does not move the funds.
- Funds move through accounts banks hold with each other.
  - **Nostro**: our account at your bank.
  - **Vostro**: your account at our bank.
- More intermediaries can add fees, delays, reconciliation work, and uncertainty, although pricing is not mechanically one fee per hop.
- Banks may prefund correspondent balances or instead use credit lines, netting, and intraday liquidity.

Nostro and Vostro T-Accounts →



Prefunding Cost and Settlement Risk

[Core Concept]

One useful economic-cost decomposition is:

$$\text{Economic cost} = \text{fee} + \text{liquidity cost} + \text{FX cost} + \text{risk charge}$$

- **Liquidity cost**: prefunded balances and committed credit consume funding and liquidity capacity. Measure their funding or opportunity cost.
- **FX settlement risk**: paying one currency before receiving the other exposes the full principal. The economic cost includes expected loss and capital or liquidity charges.
- **Payment-versus-payment (PvP)**: makes final transfer of one currency conditional on final transfer of the other, eliminating FX principal risk for covered transactions. Liquidity and operational risks remain.

Compare the fee, FX conversion, funding needs, and risk charges for the same payment amount and delivery time. Their relative importance varies by corridor.

SWIFT Messages, Tracking, and Settlement [Core Concept]

- SWIFT standardizes and securely transmits messages between financial institutions. The actual money moves on bank and settlement-system ledgers.
- Tracking services can show payment progress and improve transparency. Recipient-bank processing, compliance, FX, and operating hours can still affect delivery.
- CLS provides payment-versus-payment settlement for eligible FX trades. Its linked settlement reduces principal risk on covered trades, while liquidity and operational risks remain.
- CIPS supports RMB clearing and settlement. It can use direct and indirect participation and coexist with other messaging and correspondent arrangements.

New rails try to shorten or replace the correspondent chain:

- **CIPS**: China's payment system clears and settles eligible renminbi payments through direct and indirect participants. It can shorten some correspondent chains but also interoperates with SWIFT messaging.
- **Multi-CBDC experiments**: platforms such as mBridge test wholesale CBDC payments among participating commercial banks under multiple central banks. They are not yet universal production rails.
- **Stablecoin sandwich**: convert local currency into a dollar stablecoin, send it on a blockchain, then convert into the destination currency.

Each aims for lower fees and faster finality. Each raises new questions on regulation, reserve backing, and legal finality across borders.

Multi-CBDC and Nexus Detail →

Payment Connect and Cross-Border Retail Transfers

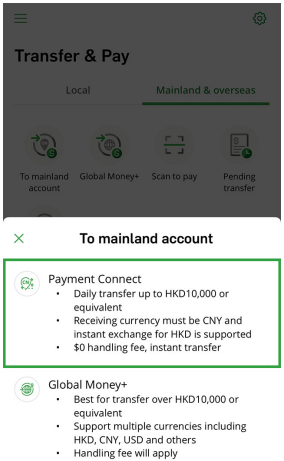
[Background Context]

- Payment Connect launched on 22 June 2025, linking mainland China's Internet Banking Payment System with Hong Kong's FPS.
- The launch supports real-time cross-boundary remittances through participating institutions, subject to rules and limits.
- A remittance linkage differs from a cross-border merchant QR acceptance scheme. Paying at a Shenzhen shop with a wallet is not an adequate description of every Payment Connect use.
- Compare recipient eligibility, amount limits, exchange rates, operating hours, and fees for the relevant corridor.

Source: HKMA and PBoC, launch announcement, 20 June 2025.

Mainland & HK Payment Connect

[Background Context]



Payment Connect links FPS and mainland IBPS for eligible cross-border remittances. It launched on 22 June 2025.

CBDCs, Smart Contracts, and Tokenized Deposits

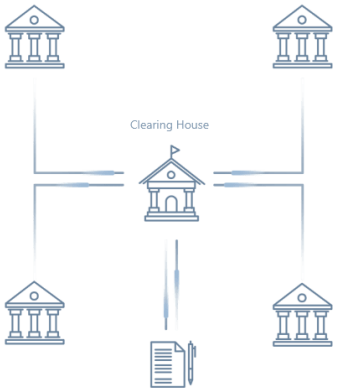
[Background Context]

- A CBDC is a digital liability of a central bank. Retail access, wholesale access, privacy, and ledger technology are design choices.
- A tokenized deposit remains a commercial-bank liability. A wallet balance and a stablecoin may represent claims on different private issuers.
- Smart contracts execute specified code. Upgrade permissions, administrator keys, external data feeds, and dispute processes determine who must still be trusted.
- For a documentary credit, code can release payment after a verified event. Reliable evidence of shipment and legally recognized performance remain necessary. Programmability is not unique to CBDCs.

Shipment escrow: worked sequence →

Centralized (CLT) vs Decentralized (DLT) Ledger Technologies

[Background Context]



Ledger architecture and governance are separate: a distributed ledger can be permissioned and centrally governed.

Centralized (CLT) vs Decentralized (DLT) Ledger Technologies

[Background Context]

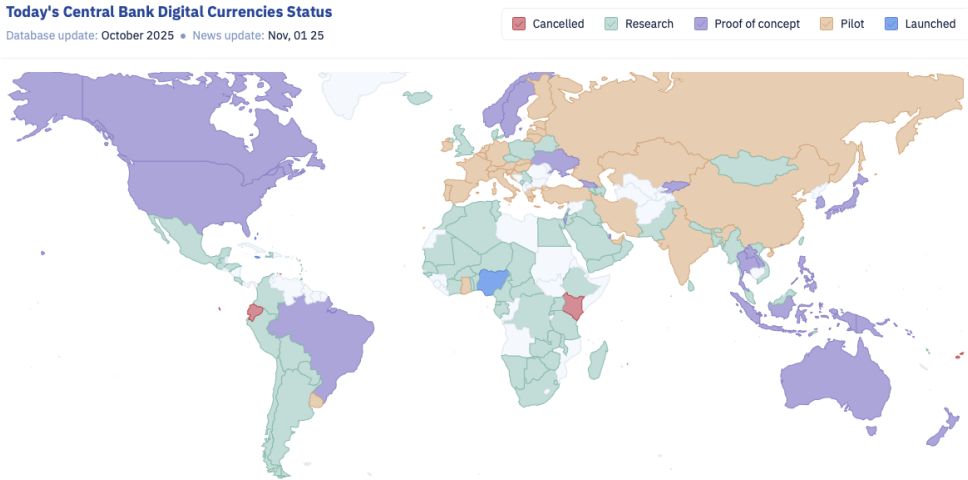


CBDC Interoperability Experiments

- Project Dunbar explored shared platforms for wholesale multi-CBDC settlement. Project Icebreaker studied a hub linking retail CBDC systems.
- mBridge explored direct payments and FX transactions among banks using participating central banks' digital currencies. A pilot or minimum viable product is not universal market coverage.
- Liquidity-saving mechanisms coordinate queued obligations to reduce prefunding. They need rules for priorities, failed payments, and settlement finality.
- Cross-border implementation also requires FX liquidity, governance, access controls, and compatibility with each jurisdiction's legal framework.

Dunbar, Icebreaker, and mBridge →

CBDCs Status around the World

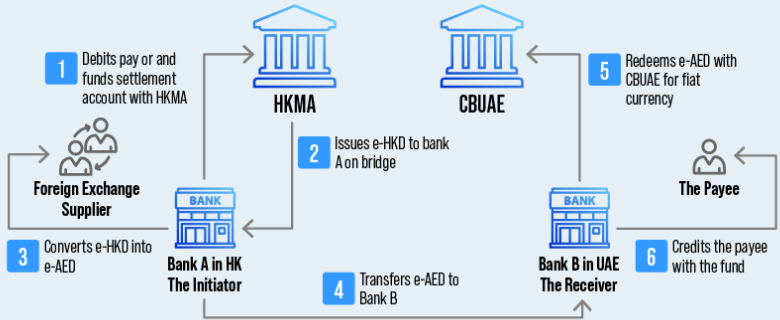


Historical CBDC-status snapshot from the 2025 materials. Research and pilot projects differ from launched currencies.

Source: CBDC Tracker.

Wholesale CBDC cross-border payment of a transaction from Hong Kong to UAE

Figure 1. The procedure of cross-border payments on bridge



Source: TABInsights

mBridge lowers funding costs and alleviates liquidity constraints on banks through liquidity saving mechanism (LSM)

Figure 2. Traditional Settlements versus LSM-enabled Settlements

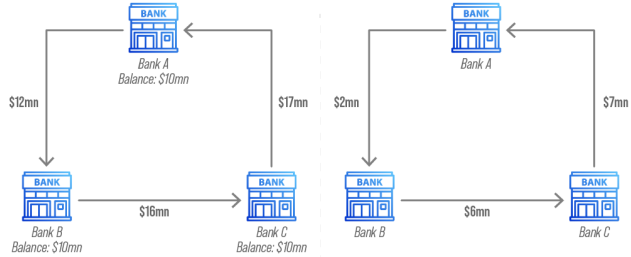


Figure 2a. Traditional Settlements

Figure 2b. LSM-enabled Settlements

Source: TABInsights

Liquidity-saving mechanisms offset or coordinate obligations to reduce funding needs while preserving settlement conditions.

Nexus and Aperta Address Different Problems

[Background Context]

- Project Nexus connects domestic instant-payment systems through common technical and operational arrangements.
- Project Aperta focuses on cross-border open-finance data connectivity through APIs. Sharing permitted account information is different from settling CBDC payments.
- A standard connection can reduce the need for repeated bilateral integration, but data consent, legal access, and participant governance remain necessary.
- Neither project implies that every domestic rail is linked or that all cross-border payments automatically settle atomically.

Source: BIS, Project Aperta.

A Model of Intermediary Holdup in FX Chains (Duffie and Wang, 2026)

- Multi-hop FX swaps (Local A → USD Stablecoin → Local B) require liquidity providers across multiple legs.
- In the model, sequential contracting can create holdup: after Leg 1 is committed, a downstream counterparty can bargain over Leg 2.
- This is a bargaining result under the paper's network and commitment assumptions, not a claim that all traditional multi-dealer trades fail.

Atomic Settlement via Smart Contracts

- **Atomic ("All-or-None") Execution**: a contract can condition every transfer it controls on approval of the entire package.
- Atomicity can remove principal and bargaining exposure within that package. It does not by itself remove prefunding, market, liquidity, oracle, or legal-finality risks—especially across independent ledgers.

Atomic FX Bargaining Detail →

The Multinational Treasurer's Payment Calendar [Core Concept]

A Hong Kong parent sells in Europe, buys components in the US, and pays staff in Hong Kong. Treasury must have funds in the right account when each payment is due.

Date	Legal entity	Cash flow	Amount
Monday	European subsidiary	Customer receipts	EUR 500,000
Wednesday	Hong Kong parent	Supplier payment	USD 300,000
Friday	Hong Kong parent	Payroll	HKD 1,200,000

- The group may have surplus cash while one subsidiary lacks the currency it needs.
- Cash becomes available at different times as customers pay and transfers settle.

Netting and the Amount That Must Cross the Border

[Core Concept]

Two subsidiaries have reciprocal invoices due on the same date: the European entity owes the Hong Kong entity EUR 500,000, and Hong Kong owes Europe EUR 320,000.

$$\text{Net payment from Europe to Hong Kong} = 500{,}000 - 320{,}000 = \text{EUR } 180{,}000.$$

- If their agreements and local rules permit netting, treasury can settle one net amount while recording both invoices as discharged.
- At an assumed 0.4% transfer cost, two gross transfers cost EUR 3,280. One net transfer costs EUR 720, saving EUR 2,560.
- Keeping EUR 180,000 for a later euro supplier bill may also avoid two FX conversions. Compare its payment date with other uses of the cash.

What a Payment Provider Does

[Background Context]

- A provider collects the payer's money, converts currencies when requested, and arranges delivery to the beneficiary. A local payout may use the provider's bank or payment-system partner in the destination country.
- **Wise Business**: batch-payment tools group multiple recipient payments and allow the business to fund the batch from an account balance or bank transfer.
- **Airwallex**: payout tools offer local and SWIFT routes where supported. Beneficiary details, eligible currencies, and fees depend on the route.
- Treasury compares the total cost, the amount the recipient receives, and the arrival date.

Source: Wise, funding batch payments.

Source: Airwallex, payout network.

Using a Provider for a Supplier Payment

[Background Context]

The supplier must receive exactly USD 100,000 by Wednesday. Treasury uses HKD cash.

1. Confirm the invoice, beneficiary name, account currency, bank details, and payment reference. Verify changes to the supplier's bank details through a known contact.
2. Request a quote with the **recipient amount fixed**. Record HKD debited, all fees, quote expiry, and the stated delivery window.
3. Approve the transfer and send funds to the provider before the cut-off.
4. Track completion and reconcile the HKD debit, FX rate, fees, and supplier receipt to the invoice. Investigate a short payment or return.

Provider Comparison: A Fixed Recipient Amount [Core Concept]

Hypothetical quotes for delivering USD 100,000. The mid-market rate is 7.80 HKD/USD. Define each FX markup as a percentage increase in that rate. All listed fees are paid separately by the sender.

	Bank A	Provider B	Provider C
FX markup	0.60%	0.25%	0.10%
Sender fee (HKD)	250	80	50
Additional guaranteed delivery fee (HKD)	150	0	0
Stated arrival	1 business day	1 business day	3 business days

1. Calculate the HKD debit for each route.
2. The supplier's deadline is two business days away. Choose the least-cost route that meets the stated deadline.

Provider Comparison: Cost and the Deadline

[Core Concept]

$$\text{HKD debit} = 100{,}000(7.80)(1 + \text{markup}) + \text{fees}.$$

Route	Conversion amount	Total HKD debit	Meets deadline?
Bank A	784,680	785,080	Yes
Provider B	781,950	782,030	Yes
Provider C	780,780	780,830	No

- B saves HKD 3,050 relative to A and meets the stated deadline. C arrives too late for this payment.
- A quote that deducts intermediary fees from the transfer can leave the supplier short. Fix the recipient amount when comparing quotes.

Delay Risk Can Change the Route Choice

[Core Concept]

Now assume A is late with probability 0.5% and B with probability 5%. A late delivery causes a known HKD 100,000 production loss. Otherwise the production loss is zero.

	Bank A	Provider B
Certain payment cost	785,080	782,030
Expected delay loss	$0.005(100{,}000) = 500$	$0.05(100{,}000) = 5{,}000$
Expected total cost	785,580	787,030

A now has the lower expected cost. B is cheaper when the loss per late payment is below

$$L^* = \frac{785{,}080 - 782{,}030}{0.05 - 0.005} = \text{HKD } 67{,}777.78.$$

Working Capital at a Payment Provider

[Core Concept]

A provider requires the firm to place USD 500,000 in its account three days before payment. The firm could otherwise earn 5% per year. Use a 360-day year and simple interest.

$$\text{Prefunding opportunity cost} = 500{,}000(0.05)\frac{3}{360} = \text{USD } 208.33.$$

- Prefunding makes cash available for the payment but removes it from other uses. If the firm borrows to prefund the account, include the interest cost.
- Check who legally holds the balance and how funds are protected. A provider balance may have different protections from a bank deposit.
- A second route and cash buffer help if an account review, technical failure, or currency cut-off delays access.

Funding an Overseas Investment Account

[Core Concept]

An investor in Hong Kong wants to buy a USD-denominated bond through an overseas broker.

1. Obtain deposit instructions for the correct broker account and currency. Include the investor identifier required for reconciliation.
2. Check whether the broker accepts the sending account and payment route. The broker must be able to match the incoming payment to the investor.
3. Compare converting HKD at the bank, at an accepted provider, or at the broker after depositing HKD. Include all charges.
4. Wait until funds are available for trading and settlement. Place the trade with enough settlement currency for price, accrued interest, and fees.

Source: IBKR, Fund Your Account.

Investor Exercise: Two Ways to Fund a Broker [Core Concept]

The investor has HKD 78,000 in total. Both hypothetical routes are accepted by the broker and deliver USD before settlement. Ignore tax and investment trading fees.

	Bank conversion	Broker conversion
Transfer fee	HKD 150, deducted first	HKD 10, deducted first
Conversion rate	7.84 HKD/USD	7.80 HKD/USD
Conversion commission	Included in rate	USD 2, deducted after conversion

1. Calculate net USD available under each route.
2. How much is saved by the cheaper route, measured in USD at these quotes?
3. Which route would you use if only the bank transfer arrived before settlement?

$$\text{Bank route} = \frac{78{,}000 - 150}{7.84} = \text{USD } 9{,}929.85.$$

$$\text{Broker route} = \frac{78{,}000 - 10}{7.80} - 2 = \text{USD } 9{,}996.72.$$

The broker route leaves USD 66.87 more to invest.

- If the broker route arrives too late, use the bank route or postpone the trade.
- FX conversion buys USD cash. Buying a USD bond then adds credit and interest-rate risk. Each transaction changes a different exposure.
- Selling the bond later creates another settlement date. Withdrawal, conversion, and the eventual HKD receipt are further steps.

Institutional Investors: Securities and Cash Settlement [Core Concept]

- A fund instructs its broker to trade and its custodian to deliver or receive securities. The custodian records the fund's holdings and cash.
- **Delivery versus payment (DvP)** links securities delivery to the matching cash payment. **Payment versus payment (PvP)** links the two currency legs of an FX trade.
- The investor must have the settlement currency available when the securities arrive. Failed settlement can create penalties, replacement trades, or lost access.
- A currency hedge may settle on another date or require collateral before the securities are sold. Asset value and immediately available cash must be planned separately.

A Cash Calendar for an International Fund [Core Concept]

Hypothetical USD cash flows, in millions. Initial available cash is 1.0. Assume no overdraft or intraday credit.

Date	Transaction	Cash flow	Running cash
Today	FX hedge collateral	−0.8	0.2
Tomorrow	New bond purchase settles	−2.0	−1.8
Day 2	Previous bond sale settles	+3.0	1.2

1. How much additional cash must arrive before tomorrow's purchase?
2. Compare an earlier FX conversion, committed credit, and postponing the purchase.

Stablecoin Conversion and Wallet Balances

[Background Context]

- In a stablecoin payment route, the sender buys a token, the token moves on a ledger, and the recipient or intermediary redeems or sells it for local money.
- Total cost includes both fiat conversion legs, spreads, blockchain fees, custody, and compliance. A quick token transfer need not mean a quick final bank payout.
- Fiat-backed and algorithmic stablecoins use different stabilization mechanisms. Reserve quality, redemption access, and legal claims matter for run risk.
- Platform wallets such as Alipay and WeChat Pay combine customer interfaces with payment services. Some transactions use wallet balances and others link to bank accounts. Safeguarding and interest-on-float rules differ by jurisdiction.

Wallet institutions and legal claims →

Payment Route Exercise

[Background Context]

Your firm must deliver **USD 1,000,000 in full within two business days**. Compare correspondent banks (A), PvP settlement (B), and a stablecoin bridge (C).

1. Understand the routes and follow one worked example.
2. Practise FX cost, committed-cash cost, the risk charge, total cost and expected cost. Hints and solutions are free.
3. Choose for the baseline and a new payment after a delay.
4. Experiment, save a comparison, then explain your choice.



Open exercise

Student explanation

20–25 minutes. Any route choice can earn participation credit. Complete practice or review its solutions, make both choices, save a comparison and submit the reflection.

Inputs and cost calculation → | Worked example | Treasury workshop

Payment Route Debrief

[Background Context]

All costs are USD above the same USD 1M payment. Baseline quotes and access are classroom assumptions.

Cost line	A: Correspondent	B: PvP	C: Stablecoin
Fixed fee	150	400	50
FX conversion	800	400	600
Committed-cash cost	200	20	5
Assumed risk charge	500	0	100
Total	1,650	820	755

- C costs $65 less than B. Its off-ramp and redemption assumptions matter. Raising its off-ramp spread from 3 to 5 bp adds $200, making B cheaper.
- A costs $2,700 after a five-day delay and misses the deadline. At 20% delay probability, its expected cost is $1,860 before production losses.

Delay calculation → Cost bars, probability curves and scenarios

Dollar Invoicing and Trade-Price Pass-Through [Core Concept]

- In the dominant-currency paradigm, firms in two non-US countries can still invoice their trade in USD.
- When the importer's currency weakens against USD while the dollar invoice stays fixed, the importer's local-currency price rises. Dollar price stickiness can strengthen this pass-through.
- Bilateral exporter-importer exchange rates alone may therefore predict trade prices poorly. Dollar working-capital debt can reinforce the invoicing choice.
- Reserve holdings, invoicing, FX turnover, and bank funding measure distinct dollar roles. FX turnover currency shares sum to 200% because every trade has two currencies.

Dollar invoice: worked example →

Five Stages of an International Transaction

Liu separates five linked services. The currency can differ across stages.

Stage	Firm's decision
Pricing	Which currency fixes the invoice and the amount owed?
Financing	Which currency does the buyer borrow before paying?
Risk transfer	Which currency exposures are hedged or insured?
Payment	Which asset and rail discharge the obligation?
Reinvestment	Does the recipient hold, invest or convert the proceeds?

A USD stablecoin bridge changes the payment mechanism while the invoice can remain in USD. Each stage can reinforce demand for the currency used at the others.

Source: Zongyuan Zoe Liu, Foreign Affairs article supplied for Week 10, printed p. 7.

Invoicing Currency and Settlement Currency

[Background Context]

- A contract can specify **USD 100,000** while the exporter receives RMB after an intermediary converts the payment at settlement.
- If the USD invoice stays fixed, the buyer still needs USD 100,000. The conversion timing determines the exporter's RMB receipt and who bears FX risk.
- Liu reports **6.5% of China's exports invoiced in RMB in 2023**, versus **nearly 30% of trade in goods settled in RMB** in the statistics she discusses.
- These are different measures and denominators: exports versus goods trade, invoicing versus settlement. Their difference is not a measure of misreporting or the share converted at settlement.

Source: Liu, supplied article, printed p. 8; citing IMF invoicing estimates and Chinese settlement statistics. The article does not specify the same reference year for the settlement share.

Payment Infrastructure and Trade Financing

[Background Context]

- **Payment reach:** Liu reports 1,829 CIPS participants in June 2026, with payments routable to over 5,200 banking institutions in 191 countries. Reach and transaction value measure different things.
- **Payment scale:** her 2025 value comparison puts CHIPS at about 19 times CIPS and T2 at about 14 times CIPS. System scope and transaction coverage differ; these ratios are not global currency shares.
- **Trade financing:** in May 2026, RMB accounted for 7% of the value of trade-finance messages on SWIFT, versus 82.5% for USD. This is a messaging-based financing measure, not all trade or all payments.

Source: Liu, supplied article, printed pp. 7 and 10–11. Figures are the article's dated reports, not a new measurement exercise.

Reinvesting Payment Proceeds

[Background Context]

- After receiving RMB, a firm can hold a deposit, buy an asset, pay another supplier, or convert the balance. Repeated willingness to hold and reinvest supports currency use.
- Liu discusses Stock Connect and Bond Connect as investment channels, and Swap Connect as a way to hedge onshore RMB interest-rate risk. Access, liquidity and the ability to move funds shape demand.
- She proposes RMB green bonds as one possible asset supporting that demand. Issuance and environmental eligibility alone do not establish credit safety, market depth or liquidity in stress.
- **Link to Week 12:** evaluate the issuer's cash flows, loss allocation, tradability and currency exposure alongside the use of proceeds.

Source: Liu, supplied article, printed pp. 11–15. The green-bond channel is an argument to assess, not an established replacement for dollar safe assets.

Dollar Pricing and Expenditure Switching

[Core Concept]

Textbook view (producer-currency pricing): each exporter sets a sticky price in its own currency, so the **terms of trade** (export prices relative to import prices) move one-for-one with the exchange rate.
Dominant-currency pricing: most exporters set sticky prices in dollars. Gita Gopinath estimates that close to 90% of China's exports to the US are priced in dollars, so the terms of trade move little.

Shock	That country's exports	That country's imports
Dollar depreciates (US)	Rise: cheaper in foreign currency	Little change: dollar price fixed
RMB depreciates (China)	Little change: dollar price fixed	Fall: cost more in renminbi

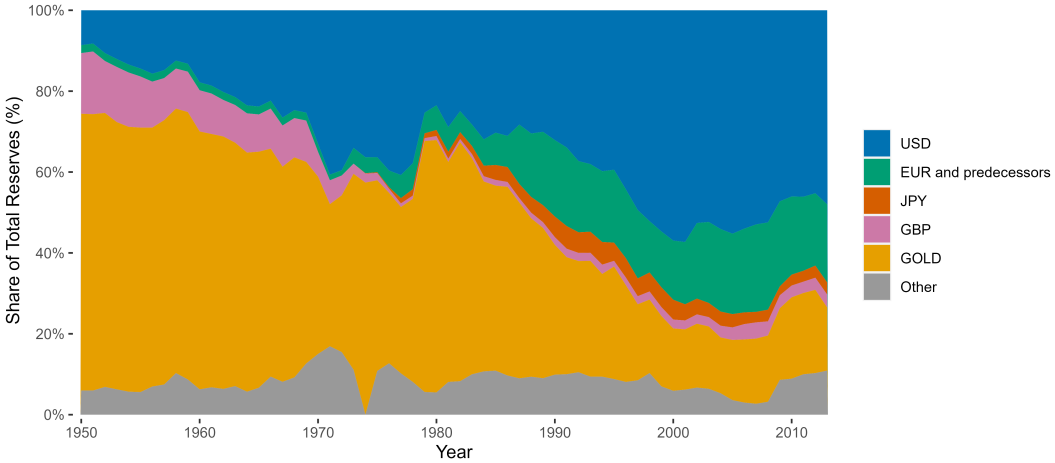
Expenditure switching: buyers shift spending toward goods that became relatively cheaper. Under dollar pricing it works through US exports but not US imports, and through other countries' imports but not their exports.

Source: Gita Gopinath, Conversations with Tyler, Ep. 291, recorded 27 August 2026.

Why prices stay in dollars → Depreciation vs. Tariff →

Time Series of the Currency Composition of Reserves

[Background Context]



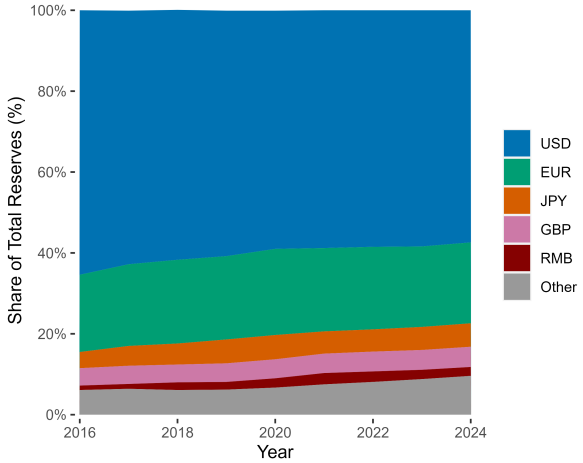
Source: My figure, data from Eichengreen, Mehl, Chitu (2017).

Trade-Weighted Strength of USD and US Monetary Policy

[Background Context]



Official Reserve Currency Shares, 2016–2024



Historical official-reserve shares. Reserve holdings and trade invoicing measure different uses of international currencies.

Source: My figure, data from Eichengreen, Mehl, Chitu (2017).

Dollar Dominance Has Several Dimensions [Core Concept]

The dollar is widely used in trade, payments, borrowing, and reserves.

Function	Why it matters to a firm or investor
Trade invoicing	Sets the currency of prices and receivables.
Payments and FX trading	A common currency can connect markets with deeper liquidity.
International borrowing	Borrowers need the currency to service debt and refinance.
Reserves and collateral	Investors value assets they can sell or pledge during stress.

These uses can reinforce each other. A new payment rail changes how a currency moves, while contracts determine which currency is owed.

Source: Federal Reserve, international role of the US dollar (2025).

The US External Balance Sheet

[Core Concept]

The US international investment position records residents' claims on the rest of the world and foreign residents' claims on the US.

- External assets include foreign direct investment, portfolio securities, loans, and deposits. Many assets have foreign-currency exposure.
- External liabilities include US equity and direct investment as well as Treasury securities, deposits, and other debt. Much US external debt is denominated in USD.
- Changes in asset prices and exchange rates alter both sides. A dollar decline raises the USD value of a fixed foreign-currency claim.
- The US external balance sheet combines the foreign assets and liabilities of households, firms, banks, and government.

Source: BEA, International Economic Accounts: Concepts and Methods.

Example: A Weaker Dollar and US Net Foreign Assets [Core Concept]

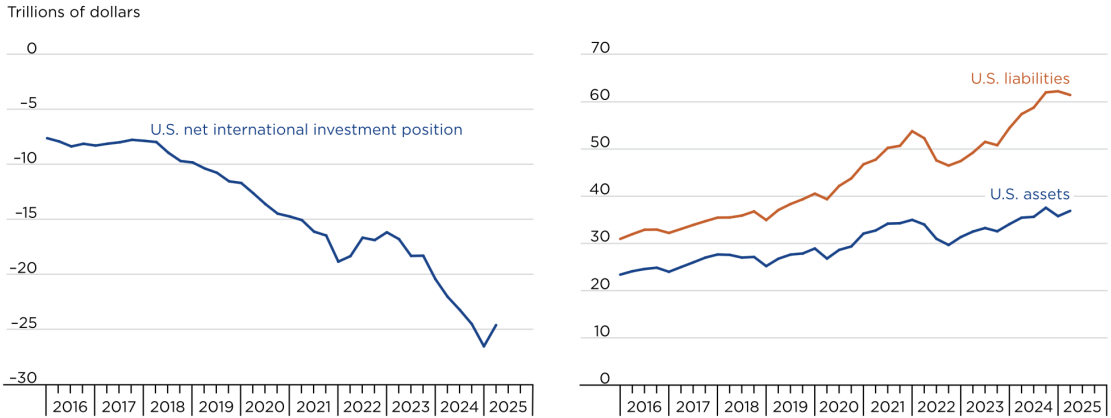
Consider US residents who own EUR 100B of foreign assets and owe USD 100B. Hold the euro asset price and dollar debt fixed.

USD billions	Initially	After the dollar weakens
Exchange rate (USD/EUR)	1.00	1.20
Foreign assets in USD	100	120
Dollar liabilities	100	100
Net foreign assets	0	20

Discuss: Who gains from the exchange-rate change? What happens to a euro investor who holds the USD 100B claim?

US External Position: Historical Evidence

[Background Context]



U.S. Bureau of Economic Analysis

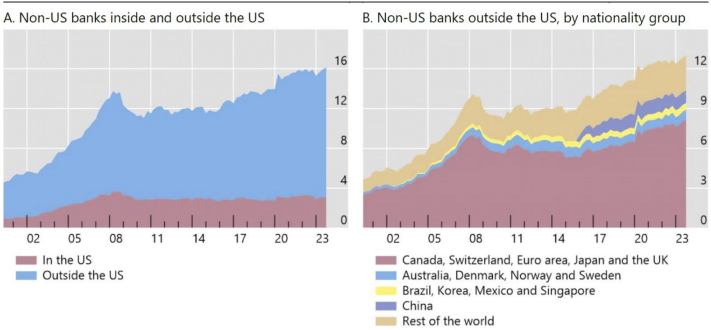
Historical chart from the 2025 course deck, based on BEA international investment position data. Asset-price and currency changes can move the position even without new borrowing.

Source: BEA, international investment position.

Dollar Liabilities outside the United States

[Background Context]

Figure 2: Dollar Liabilities of Banks Headquartered outside the United States



Note: Data depict cross-border and local liabilities in all instruments vis à vis counterparty countries. Data exclude intragroup positions but include liabilities to other (unaffiliated) banks. From the end of 2015, data include positions reported by China and Russia (the latter up to the end of 2021). Data are in trillions of dollars.
Source: BIS consolidated banking statistics, BIS local banking statistics, and author's calculations

Historical figure from McCauley (2024), reproduced in the 2025 course deck. Offshore dollar claims explain why non-US banks also need dollar liquidity.

Valuation Changes and the Current Account

[Core Concept]

In this accounting example, changes in net foreign assets come from the current account and valuation. All amounts are USD billions.

$$\Delta NFA = \text{current account balance} + \text{valuation changes}.$$

Opening net foreign assets	−500
Current account deficit	−100
Net gain from asset prices and exchange rates	+150
Closing net foreign assets	−450

The external position improves by 50 despite a current account deficit of 100.

Source: BEA, international accounts methods.

Exorbitant Privilege and Exorbitant Duty

[Core Concept]

- **Funding advantage**: investors may accept a lower yield for the liquidity and collateral services of dollar assets.
- **Return differential**: the US may earn a different average return on external assets than it pays on liabilities. Differences in risk and valuation affect this comparison.
- **Exorbitant duty**: when risky foreign assets fall and safe dollar liabilities hold their value, the US external portfolio can lose wealth during global stress.
- Tabova and Warnock (2025) find that the positive differential in international *portfolio* returns disappeared in their data. Dollar assets can still provide liquidity and collateral services.

Source: Tabova and Warnock (2025), Exorbitant Changes in Three Parts.

Example: What Is a Funding Advantage Worth? [Core Concept]

Suppose investors accept 4.0% on a liquid dollar bond, versus 4.5% on an otherwise comparable bond. Each issuer borrows USD 100B for one year.

$$\text{Annual interest saving} = 100\text{B}\,(4.5\% - 4.0\%) = \text{USD } 500\text{M}.$$

Foreign asset losses: The issuer invests the USD 100B in foreign equities. Their local price falls 20%, and their currency falls 10% against USD.

$$\text{Ending asset value} = 100(0.80)(0.90) = \text{USD } 72\text{B}.$$

Discuss: Compare the USD 500M funding saving with the USD 28B asset loss. What additional cash is needed for interest?

Global Dollar Liquidity and the Federal Reserve [Core Concept]

- A non-US bank can own long-term dollar assets while funding them with short-term dollar borrowing or FX swaps. It must find dollars when that funding matures.
- If private lenders stop rolling over claims, selling assets can depress prices and weaken other banks holding the same assets.
- A Fed swap line supplies dollars to a partner central bank, which lends them to eligible banks against its own terms and collateral.
- The partner central bank bears the credit risk on its loans to banks. Dollar liquidity can bridge a funding shortage, while losses on bad assets still require capital or restructuring.

Source: BIS, foreign-currency funding measures (2008).

Safe-Asset Demand and the New Triffin Dilemma

[Core Concept]

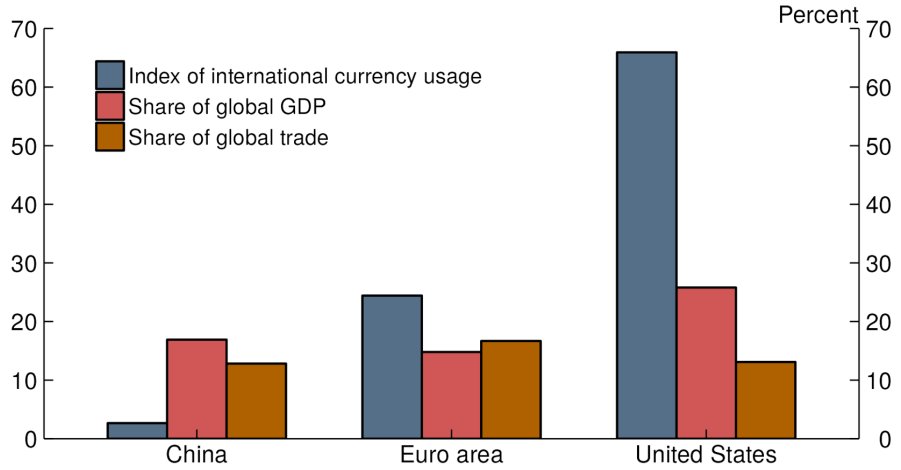
World demand for liquid dollar claims can grow faster than the issuer's tax base. More debt supplies reserves and collateral, while increasing future interest payments.

Illustration: Debt grows from USD 20T to 25T; annual revenue stays at USD 5T. Assume all debt is financed at the yield shown.

	Initial	More debt	More debt, higher yield
Debt (USD T)	20	25	25
Yield	3%	3%	4%
Annual interest (USD T)	0.60	0.75	1.00
Interest / revenue	12%	15%	20%

Discuss: How could doubts about repayment raise the cost of supplying safe assets?

Economic Size and International Currency Use



2022 observations, from the 2025 course deck. Economic size and currency use have different denominators.

Source: Federal Reserve, RMB internationalization (2024).

What Would Support Greater Euro or Renminbi Use?

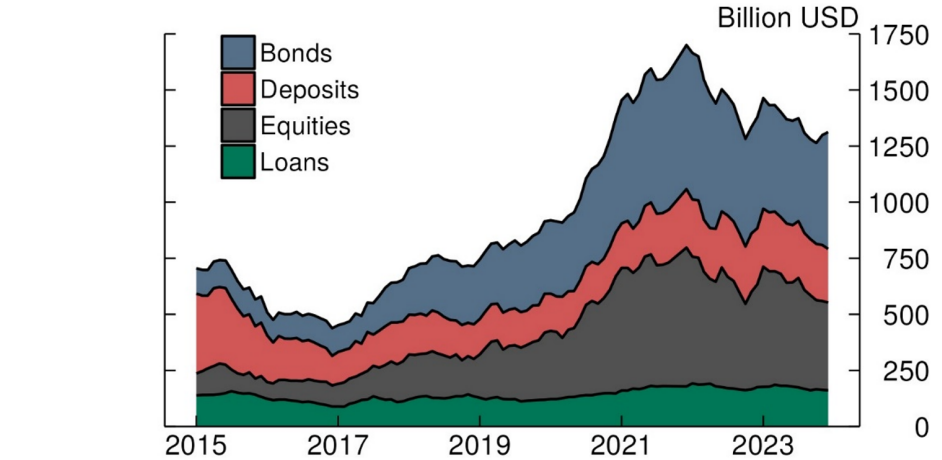
Consider the same four tasks for any proposed currency alternative.

1. Can firms invoice customers and reliably obtain the currency for debt service?
2. Can investors enter, hedge, and exit large positions at reasonable cost?
3. Is there a deep supply of liquid assets usable as collateral?
4. Will institutions support settlement and emergency liquidity during stress?

Euro markets face differences across sovereign issuers. Renminbi use depends partly on cross-border access and capital-account restrictions. Greater international use also requires deep asset markets and reliable access to funding.

Source: Federal Reserve, RMB internationalization (2024).

Foreign Holdings of Chinese Assets



Historical observations, from the 2025 course deck. Capital-account rules shape foreign access to Chinese bonds and equities.

Source: Federal Reserve, RMB internationalization (2024).

Exorbitant Privilege and Borrowing Costs

[Background Context]

- Investors hold dollar assets to make payments, store cash, and provide collateral.
- Demand for safe, liquid dollar claims can raise their prices and lower their yields.
- This funding advantage is part of **exorbitant privilege**: the US can borrow in its own currency on favorable terms.
- Firms may benefit through lower benchmark yields. Their borrowing costs also depend on credit risk, maturity, and hedging.

If the advantage shrinks

Holding other drivers fixed, investors require higher yields. Existing bondholders lose value, and borrowers pay more on new or refinanced debt.

Source: Background: Federal Reserve, The International Role of the US Dollar (2025).

A Fall in Demand for Dollar Assets

[Core Concept]

Scenario: investors reduce the share of their portfolios held in dollar assets.

- **Currency**: selling dollars for other currencies puts downward pressure on the dollar.
- **Bonds**: selling Treasuries puts downward pressure on their prices and upward pressure on their yields.
- **Funding**: higher market yields can raise the cost of issuing debt or refinancing a loan.
- **What investors buy matters**: a switch from long Treasuries to dollar bills mainly changes maturity exposure. A switch into euro assets also changes currency exposure.

Working assumption for the examples

The dollar weakens and dollar yields rise. Other shocks, including central-bank rate cuts, can change the outcome.

A Weaker Dollar: Who Gains and Who Loses? [Core Concept]

Assume fixed dollar amounts and an appreciation of the firm's home currency.

Exposure	Immediate effect	Decision
Dollar sales receipts	Convert into less home currency	Hedge net receipts or revisit pricing
Dollar import payments	Cost less in home currency	Reassess purchasing costs and hedges
Dollar debt	Principal costs less in home currency	Check interest resets and refinancing
Dollar investments	Lose home-currency value through FX	Review currency hedge and asset risk

Net exposure matters: dollar receipts can offset dollar payments due at the same time. Existing hedges change the result.

Firm Example: Dollar Revenue and RMB Costs

Hypothetical exporter: in three months, it receives USD 10m from customers and pays USD 4m to suppliers. Its other costs are in RMB.

- Net dollar receipt: USD 10m − 4m = 6m.
- At 7.20 RMB per USD, the receipt is worth RMB 43.20m.
- At 6.60 RMB per USD, it is worth RMB 39.60m: a **RMB 3.60m loss** before hedging.
- Selling USD 6m forward locks the RMB value at the quoted forward rate. Hedging USD 10m would leave the supplier payment exposed.

Treasurer's decision

Match dollar inflows and outflows by date, then hedge the remaining exposure. If sales are uncertain, compare a partial hedge with an option and its premium.

Firm Example: A Smaller Funding Advantage

Hypothetical USD 20m refinancing, with annual interest and unchanged credit spread:

	Before	After
Dollar benchmark yield	4.0%	5.0%
Firm's credit spread	2.0%	2.0%
All-in borrowing rate	6.0%	7.0%
Annual dollar interest	USD 1.20m	USD 1.40m

- Annual interest rises by USD 200,000. A wider credit spread would add to the increase.
- Existing fixed-rate debt keeps its coupon until maturity. Floating-rate debt changes when its reference rate resets.

Decision: compare refinancing dates, fixed versus floating rates, and the return on projects that need new financing.

Choosing the Currency of a Loan

[Core Concept]

- **Dollar revenue**: dollar debt can match cash inflows with debt service. Check that amounts and payment dates line up.
- **RMB revenue**: compare an RMB loan with a dollar loan hedged into RMB over the same maturity.
- Include the loan spread, FX hedge price, fees, and any collateral or rollover needs in the comparison.
- If dollar funding loses its cost advantage, local-currency borrowing may become more attractive. Use current all-in quotes to decide.

Connection to Weeks 4 and 8

The cheapest headline interest rate may come with expensive hedging. Leaving dollar debt unhedged creates a currency position that can gain or lose.

Hedge Fund: Matching a Trade to the Shock

Conditional trade ideas: each position needs a forecast relative to today's price.

View	Possible position	Main risk
Dollar weakens against EUR	Sell USD and buy EUR, or use an FX option	Dollar strengthens or option expires
Long Treasury yields rise	Reduce bond duration or short Treasury futures	Yields fall and bond prices rise
US assets remain attractive, but USD weakens	Keep assets and hedge dollar exposure	Asset prices fall despite the FX hedge

Financing matters: forward prices reflect interest-rate differences. Futures require margin. Options require a premium.

Fund Example: Bond Risk Plus Currency Risk

[Core Concept]

Hypothetical euro-based fund: USD 10m of Treasuries, modified duration 5, with no currency hedge. Ignore coupon income and financing over the shock window.

- Treasury yield rises by 1 percentage point. Approximate bond-price change: $-5 \times 0.01 = -5\%$.
- The euro value of one dollar falls by 8%.
- Euro portfolio return: $(1 - 0.05)(1 - 0.08) - 1 = \mathbf{-12.6}\%$.
- An FX hedge addresses the currency exposure. Shorter duration addresses sensitivity to yields. A hedge sized before the bond-price change may need adjustment.

Portfolio decision

Assess currency and duration separately. A dollar bond can lose through both channels at once.

A Dollar View Needs a Price and a Time Horizon

- A gradual decline in dollar demand may already be reflected in exchange rates and bond yields.
- A profitable position needs a further move relative to its entry price, after financing, hedging, and transaction costs.
- State the expected catalyst and holding period. Carry costs and margin calls can force an exit before a long-run view pays off.
- Use separate scenarios for lower dollar demand and a global scramble for dollar cash. The latter can strengthen the dollar and hurt a short-dollar position.

Fund manager's decision

Set the position size and cash buffer using the adverse scenario as well as the expected return.

April 2025: Investors Hedged Dollar Exposure

[Background Context]

- BIS analysis links part of the dollar's April 2025 fall to investors increasing hedges on their dollar assets.
- Investors can keep US bonds or equities while selling the associated dollar exposure forward.
- Dealers managing those forward positions can transmit the hedging pressure to the spot FX market.

Investment decision

The decision to own a US asset and the decision to hold its currency exposure can change separately.

Source: Shin, Wooldridge and Xia, US dollar's slide in April 2025: the role of FX hedging, BIS Bulletin 105 (20 June 2025).

Dollar Stablecoins and Currency Exposure

[Core Concept]

- A token designed to track USD 1 gives its holder dollar exposure. If the dollar falls against RMB, the token's RMB value falls even if its peg holds.
- An exporter receiving a dollar token still needs to manage the currency of its receipts and costs.
- If token issuers buy additional Treasury bills to meet new demand, this can support bill prices. Users switching from existing dollar assets can offset that demand.

Connection to the payment-routing game

Choose the payment route for its total cost and reliability. Choose the currency and hedge for the underlying cash flow.

Payment volume and token balances →

Cash Available When the Firm Needs It

[Core Concept]

- **Bank or currency change**: check whether customers can pay, suppliers accept the currency, and balances can be converted when bills are due.
- **Stablecoin route**: check who can redeem, how long redemption takes, and the fees for conversion into bank money.
- **Fund collateral**: allow for margin calls if bond prices fall or an FX position moves against the fund.
- **Contingency**: keep accessible cash and an alternative payment route for obligations that cannot wait.

Liquidity decision

A balance that holds its quoted value can still be unavailable when a payment is due.

Redemption example →

Hypothetical RMB-based firm, over the coming year:

- USD 10m customer receipts and USD 4m supplier payments arrive at year-end. Remaining operating costs are in RMB.
- USD 20m of debt is refinanced today for one year. Its annual rate rises from 6% to 7%.
- The year-end exchange rate is 6.60 RMB per USD, compared with a planning rate of 7.20. Assume the dollar cash flows stay fixed.

Discuss:

1. How much RMB value does the firm lose on its net operating receipt?
2. Does the weaker dollar offset the higher interest rate?
3. What should the treasurer compare before choosing the next loan and hedge?

Class Decision: Cash Flows and Actions

- **Operating receipt**: USD 6m falls from RMB 43.20m to RMB 39.60m. Loss: RMB 3.60m.
- **Annual interest**: the old plan was $20\text{m} \times 6\% \times 7.20 = \text{RMB } 8.64\text{m}$. The new outcome is $20\text{m} \times 7\% \times 6.60 = \text{RMB } 9.24\text{m}$.
- Interest costs rise by RMB 0.60m despite the weaker dollar. Cash after operating flows and interest falls by **RMB 4.20m**, before other costs and principal repayment.
- At year-end the USD 20m principal also falls in RMB value, from 144m to 132m. Its repayment or refinancing needs a separate funding plan.

Action: compare a dollar loan matched to receipts with a hedged dollar loan and an RMB loan. Include interest and principal in the dated exposure schedule.

A Dollar Shock and Asset Prices: The Joint Distribution [Core Concept]

A euro investor holds a USD asset. Let g be the change in EUR per USD and r the asset's USD return over one shock window. Ignore income, costs, and financing.

$$R_{\mathrm{EUR}} = (1 + r)(1 + g) - 1 = r + g + rg.$$

State	Probability	USD FX g	Treasury r	US equity r
Calm	60%	0%	1%	8%
Global flight to safety	25%	10%	5%	−20%
Loss of confidence in USD assets	15%	−10%	−8%	−15%

Use these hypothetical probabilities for the following exercises.

Exercise: Returns in the Investor's Currency [Core Concept]

Use the preceding three-state distribution. Initial wealth is EUR 1M, invested entirely in either the Treasury portfolio or the US equity portfolio.

1. Calculate the EUR return on each portfolio in every state.
2. Calculate each portfolio's expected EUR return and probability of a loss.
3. For a portfolio with initial weights of 60% Treasuries and 40% equities, calculate the return in each state and its expected return. Assume no interim rebalancing.
4. Which shock makes both the dollar and Treasury prices fall? Why would a portfolio built only around flight to safety miss this risk?

Solution: Dollar and Asset Returns Compound [Core Concept]

EUR return	Treasuries	Equities	60/40 portfolio
Calm	1.0%	8.0%	3.8%
Flight to safety	15.5%	−12.0%	4.5%
USD confidence loss	−17.2%	−23.5%	−19.72%
Expected return	1.895%	−1.725%	0.447%
Probability of a loss	15%	40%	15%

$$E[R_{T,\mathrm{EUR}}] = 0.60(1\%) + 0.25(15.5\%) + 0.15(-17.2\%) = 1.895\%.$$

The 60/40 portfolio has expected ending wealth EUR 1,004,470. Its confidence-loss outcome is EUR 802,800.

Why Averaging the Two Returns First Fails [Core Concept]

The exact expected home-currency return includes the interaction of asset and FX returns.

$$E[R_{\mathrm{EUR}}] = E[r] + E[g] + E[rg].$$

For the Treasury portfolio:

$$E[r] = 0.65\%, \qquad E[g] = 1.00\%, \qquad E[rg] = 0.245\%.$$

$$E[R_{\mathrm{EUR}}] = 0.65\% + 1.00\% + 0.245\% = 1.895\%.$$

- Multiplying mean gross returns gives $1.0065(1.01) - 1 = 1.6565\%$, which misses their covariance.
- The difference is $\mathrm{Cov}(r, g) = 0.2385$ percentage points in expected return. Treasuries and USD rise together in the flight-to-safety state.

Exercise: A Hedge on the Initial USD Principal [Core Concept]

Keep the Treasury portfolio and the same distribution. Sell the initial USD principal forward. Assume the forward rate equals the initial spot rate and the contract settles at the end. Ignore fees, collateral, and default.

$$R_{\text{hedged}} = r + rg, \qquad \text{forward payoff / initial EUR wealth} = -g.$$

1. Calculate the hedged return in each state and its expected return.
2. In which state must the investor pay cash on the forward? How much for an initial EUR 1M investment?
3. Compare the hedged and unhedged portfolios' worst state. Why does the hedge leave some FX exposure?

Solution: The Hedge Changes Both Wealth and Cash Needs [Core Concept]

State	Unhedged	Forward contribution	Hedged
Calm	1.0%	0%	1.0%
Flight to safety	15.5%	−10%	5.5%
USD confidence loss	−17.2%	10%	−7.2%
Expected return	1.895%	−1.0%	0.895%

- In flight to safety, the forward requires EUR 100,000. The Treasury holding has gained value, but the cash must be available on the settlement date.
- The confidence-loss outcome improves from −17.2% to −7.2%. The initial-principal hedge leaves the term rg exposed.

Exercise: Exorbitant Duty in a Two-State Model [Core Concept]

A simplified US external portfolio holds foreign equities worth USD 100B and owes fixed USD liabilities worth 80B. Liabilities keep their USD value. There are no new financial flows, income payments, or other positions.

State	Probability	Foreign equity local return	USD per foreign currency
Expansion	80%	10%	Rises 5%
Global crisis	20%	−30%	Falls 10%

1. Initial net foreign assets are USD 20B. Find ending net foreign assets in both states and their expected value.
2. Explain how a positive average valuation change can coexist with a large loss in the crisis state.

Solution: Average Gains and Crisis Losses [Core Concept]

USD billions	Expansion	Global crisis	Expected value
External asset value	$100(1.10)(1.05) = 115.5$	$100(0.70)(0.90) = 63$	105.0
External liabilities	80	80	80
Net foreign assets	35.5	-17.0	25.0
Change from initial 20	15.5	-37.0	5.0

- Expected valuation gain is USD 5B, but the global crisis produces a USD 37B loss. The foreign holders of the assumed fixed dollar claims retain their dollar value.
- US equity and direct-investment liabilities also change value, so a full calculation includes valuation changes on both sides.

A Treasury Decision under Correlated Dollar Shocks [Core Concept]

A firm with RMB costs has USD 6M net export receipts after dollar supplier payments and refinances USD 20M debt. The table gives three possible outcomes for annual cash after interest. The debt principal is refinanced.

State	Probability	RMB/USD	New USD rate	RMB cash after interest
Calm	60%	7.2	6%	$(6-1.2)7.2=34.56$M
Dollar demand loss	25%	6.6	7%	$(6-1.4)6.6=30.36$M
Global funding squeeze	15%	7.8	9%	$(6-1.8)7.8=32.76$M

1. Calculate expected cash and the probability that cash falls below RMB 32M.
2. Which exposure would a forward hedge cover? Which exposure requires a rate or refinancing decision?

Treasury Distribution: Expected Cash and the Shortfall [Core Concept]

$$E[\text{RMB cash}] = 0.60(34.56) + 0.25(30.36) + 0.15(32.76) = 33.24\text{M}.$$

State	Cash relative to RMB 32M requirement	Probability
Calm	Surplus 2.56M	60%
Dollar demand loss	Shortfall 1.64M	25%
Global funding squeeze	Surplus 0.76M	15%

- The expected cash exceeds 32M, but the firm still has a 25% shortfall probability.
- An FX hedge fixes the conversion of a specified dollar amount. A fixed-rate loan or interest-rate hedge addresses interest changes. Treasury must also arrange the refinancing.

- **Payments**: compare fees, conversion costs, prefunding, and the risk of delay or loss.
- **Firms**: map net cash flows by currency and date. Reassess margins, refinancing, and hedges after a dollar shock.
- **Exorbitant privilege**: a smaller funding advantage can mean higher required yields. Existing bondholders and new borrowers face different consequences.
- **Funds**: separate currency, interest-rate, and funding risk. Judge a trade against today's price and the cash needed to survive adverse moves.
- **Digital money**: changing the payment technology can leave the currency exposure in place.

Next week: emerging-market crises, sudden stops, and sovereign debt.

Appendix: The Ledger Economics Trilemma

[Background Context]

Abadi and Brunnermeier (2022) derive a trilemma under their model's incentive and enforcement assumptions.

1. **Self-sufficient (trustless)**: needs no outside power to punish record-keepers.
2. **Rent-free**: pays no rents to record-keepers.
3. **Resource-efficient**: wastes no resources such as computing power.

Where real systems land

- Bitcoin's proof of work pays validators and consumes substantial computing resources. Ethereum now uses proof of stake, with different incentive and governance costs.
- Commercial-bank ledgers can be resource-efficient, but charge fees and require trust, supervision, and operational resilience.
- A central bank ledger can be efficient and low-rent, but requires trust in the central bank.

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Appendix: Domestic Fast Payment Systems in Detail

- **Brazil PIX** (2020): the central bank operates the scheme. Broad mandatory participation by large institutions helped reach, while end-user pricing varies by user and provider.
- **India UPI**: NPCI operates an interoperable instant-payment scheme across regulated banks and apps. High usage reflects both common standards and a large mobile ecosystem.
- **US ACH and FedNow**: ACH supports batch and same-day services but is not inherently a one-to-two-day rail. FedNow (2023) offers 24x7 instant settlement among participating institutions.
- **Hong Kong FPS** (2018): supports aliases and QR payments, with selected cross-border links. Domestic interoperability does not automatically imply universal cross-border reach.

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Appendix: Nostro and Vostro Accounts

A Hong Kong bank pays a beneficiary bank abroad through a US correspondent. The same account has two names depending on whose books you read.

HK Bank's books

- **Nostro asset**: USD balance held at the US correspondent, +$1,000,000.
- In this example the balance is prefunded. In practice the bank may also use netting or a credit line.

US Correspondent's books

- **Vostro liability**: the HK bank's account on its books, +$1,000,000.
- On instruction, it debits the vostro account and credits the beneficiary bank.

The economic cost is the funding or opportunity-cost spread on the required balance. A nostro may earn interest or support many payments rather than literally sit idle.

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Appendix: Payment Route Costs with Units

Principal: USD 1M. Carry: \$100 per day per \$1M committed. One bp is 0.01%. All charges are paid separately by the sender.

Cost (USD)	A	B	C
Fixed fee	150	400	50
FX spread cost	$1\text{M}(0.0008)$	$1\text{M}(0.0004)$	$1\text{M}(0.0003 + 0.0003)$
Committed cash (USD M)	1	0.2	1
Carry cost	$1(2)(100) = 200$	$0.2(1)(100) = 20$	$1(0.05)(100) = 5$
Risk charge	$1\text{M}(0.00025)(2) = 500$	0 (PvP)	$1\text{M}(0.0001) = 100$
Total	1,650	820	755

FX cost = principal $\times$ spread. Carry = committed USD M $\times$ days $\times$\$100. C uses two 3 bp legs on the same principal (a small-spread approximation), a 1 bp redemption charge and about one hour of committed cash.

Risk charges are assumed prices, not default probabilities or realised losses. Prefunding time and settlement gap are separate inputs.

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Appendix: Multi-CBDC Platforms and Project Nexus

- **Wholesale versus retail CBDC**: wholesale CBDC settles between banks. Retail CBDC is held by households and firms.
- **mBridge**: an experimental shared platform on which central banks issue wholesale CBDC and participating commercial banks test cross-border payments and FX. Reaching an MVP is not the same as broad production readiness.
- **Project Nexus**: a governance and technical blueprint for interlinking participating domestic instant-payment systems through standardized connections.
- **Goal**: reduce cost and delay while preserving compliance and legal finality. PvP, broad access, and reduced correspondent use depend on the implemented design.

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Appendix: Atomic FX Bargaining and Holdup Elimination

Network Bargaining Mechanics (Duffie and Wang, 2026)

- Intermediaries in multi-hop networks ($G = (N, E)$) bargain bilaterally over non-cash terms (asset amounts) and cash transfers.
- Without atomicity, an intermediary completing Leg 1 first has a lower disagreement payoff on Leg 2 $\implies$ downstream counterparties extract hold-up rents.

The Greenlight Protocol & Atomicity

- Participants submit conditional "greenlights" for proposed bilateral contract terms.
- A smart contract automatically converts all greenlit proposals into binding contracts **if and only if** every node greenlights simultaneously ("all-or-none").
- **Model result**: under the paper's information, bargaining, commitment, and Markov-strategy assumptions, the protocol eliminates the modeled holdup and implements its efficient allocation. This does not establish efficiency for every real payment network.

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Appendix: Stablecoin Redemption and Cash Needs

[Background Context]

Hypothetical firm: USD 1m of dollar tokens and a USD 1m supplier payment due today.

- The firm cannot redeem directly with the issuer. It can sell tokens on an exchange at USD 0.99 each.
- Selling today raises USD 990,000 before fees, leaving a USD 10,000 shortfall.
- An eligible intermediary may redeem at par, but its fees, capacity, and settlement delay determine what it offers the firm.
- Waiting for the peg to recover leaves the supplier payment unfunded today. The firm needs another cash balance or a credit line.

Treasurer's decision

Check the redemption rights available to your own firm and the time needed to obtain spendable bank money.

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US Payment Rails: Separate Operator, Rules, and Settlement

[Background Context]

- **ACH:** batch-based credit and debit transfers, with same-day options. The Federal Reserve's FedACH and The Clearing House's EPN are operators. Nacha administers network operating rules.
- **Fedwire:** a Federal Reserve real-time gross settlement service for eligible institutions. It is commonly used for large-value, time-critical transfers.
- **RTP and FedNow:** The Clearing House launched RTP in 2017. The Federal Reserve launched FedNow in July 2023. These support instant interbank payments around the clock.
- **Customer experience versus infrastructure:** a banking app is the interface. The underlying rail determines message handling, availability, settlement, and participant obligations.

Faster customer access does not, by itself, identify whether a system settles each obligation gross, nets obligations, or uses prefunded settlement arrangements.

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Brazil PIX: Adoption through Common Infrastructure

[Background Context]

- PIX launched in November 2020 under Banco Central do Brasil. Participation was mandatory for institutions with more than 500,000 active customer transaction accounts.
- Users can initiate transfers with aliases or QR codes through participating institutions. A common standard makes it easier for users of different providers to pay each other.
- A 2022 BIS review reported a 2021 merchant cost of about 0.22% for PIX, compared with 2.2% for credit cards. These are dated reported averages, not guaranteed fees for every transaction.
- Illustration at those rates: on a BRL 1,000 sale, charges would be BRL 2.20 and BRL 22.00, a BRL 19.80 difference. Credit, chargebacks, and other bundled services also differ.

Source: BIS Bulletin 52, 2022.

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Hong Kong FPS and India UPI: Shared Access, Different Designs

[Background Context]

Hong Kong FPS

Launched in 2018, FPS supports HKD and RMB payments through participating banks and stored-value facilities. Registered phone numbers, email addresses, FPS IDs, and QR codes simplify addressing. The FPS–PromptPay QR link launched on 4 December 2023 for participating Hong Kong and Thai users and merchants.

India UPI

NPCI's UPI connects participating banks and payment apps under India's regulatory framework. Apps such as PhonePe, Google Pay, and Paytm provide user interfaces. They do not each constitute a separate interbank settlement currency. Common addressing and QR standards support network effects.

Adoption reflects access, pricing, acceptance, trust, and policy. An increase in usage alone does not identify how much growth was caused by interoperability.

Source: HKMA, November 2023 briefing.

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Smart-Contract Escrow: A Shipment Example

Illustration: an importer agrees to pay 100 token units when an exporter supplies an accepted shipping document.

1. The buyer places 100 units in an escrow contract. Funds are locked under agreed release and refund rules.
2. An authorized document provider sends an authenticated shipping event. The contract checks the required digital conditions.
3. If conditions hold, the contract transfers 100 to the exporter. Otherwise funds remain locked or follow the agreed dispute/timeout process.
4. The code cannot independently inspect physical goods. False documents, a compromised data provider, legal disputes, or flawed code can still cause losses.

Compare with an LC: both can make payment conditional on documents, but the identity of the obligor, prefunding, legal undertaking, and dispute resolution differ. Read and write permissions depend on the ledger design. A distributed ledger need not disclose every transaction to everyone.

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Cross-Border CBDC Experiments: The Original Cases

Project	Participants in the cited stage	What was tested
Dunbar (2022)	Australia, Malaysia, Singapore, South Africa	Prototypes for shared settlement using multiple wholesale CBDCs.
Icebreaker (2023)	Israel, Norway, Sweden	A hub connecting domestic retail-CBDC systems, with FX providers.
mBridge (2024)	China, Hong Kong, Thailand, UAE. Saudi Arabia joined in 2024	A shared DLT platform reached minimum viable product stage. BIS handed the project to partners in October 2024.

These are dated project milestones. Technical feasibility does not establish universal access, deployment at scale, or harmonized law and compliance.

Source: BIS, mBridge.

Source: BIS, Dunbar.

Source: BIS, Icebreaker.

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Wallets and Stablecoins: Trace the User's Claim

[Background Context]

- **Platform wallets:** Alipay developed around Alibaba's commerce ecosystem. WeChat Pay around Tencent's social platform. Payments, commerce, and messaging can lower customer-acquisition costs and reinforce network effects.
- A wallet may hold stored value, display a linked bank deposit, or initiate a card payment. Apple Pay, a stored-value wallet, and a bank app need not create the same legal claim.
- A stablecoin purchase exchanges bank money for a claim or token under the issuer's redemption terms. A later redemption reverses that step. Secondary-market trading need not involve the issuer.
- Reserve investment income, transaction fees, and related financial services are possible revenue sources. Reserve segregation and regulation determine who can use balances and who bears losses.

Always ask: who owes the user money, what backs that obligation, and where does final settlement occur?

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Payment Route: Isolate the Shock and Recompute

Baseline costs are Route A \$1,650 and Route B \$820. For Route A, the scenario increases prefunding from 2 to 5 days and the settlement gap from 2 to 5 days. Fees and FX spread are unchanged.

1. Extra prefunding cost: $(5-2)$ days $\times$ \$100/day $=$ \$300.
2. Extra settlement-risk charge: \$1M $\times$ 0.00025 $\times (5-2) =$ \$750.
3. New Route A total: $1{,}650 + 300 + 750 =$ \$2,700.
4. If Route B remains \$820, its advantage grows from \$830 to \$1,880. Check whether the same shock would also change Route B's assumptions.

Decision rule: compare the complete expected cost, time, finality, and risk of each feasible route. A low advertised fee need not imply a low all-in cost.

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Dollar Invoicing: The Buyer's Currency Bill Changes

[Background Context]

A Vietnamese exporter invoices a Korean customer USD 100. The dollar invoice price is fixed for this order. The buyer has no hedge.

1. At 1,300 KRW per USD, the buyer's cost is $100 \times 1{,}300 =$ KRW 130,000.
2. At 1,430 KRW per USD, the same invoice costs KRW 143,000, an increase of KRW 13,000 or 10%.
3. The buyer may reduce future demand or absorb the increase in its margin. Immediate pass-through depends on contracts, inventories, and local pricing.
4. The exporter still receives USD 100. Its VND proceeds depend on USD/VND and its own hedges and input costs.

Dollar invoicing can transmit dollar exchange-rate changes between two non-US trading partners. Invoice currency, payment currency, and producer location are separate concepts.

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Stablecoin Payment Flows Do Not Equal Reserve Stocks

[Background Context]

Hypothetical annual trade flow: CNY 2 trillion. Suppose 20% is paid through a stablecoin whose average balance turns over 20 times per year.

1. Annual stablecoin payment flow is $2\text{T} \times 20\% = \text{CNY } 400\text{B}$.
2. From $V = vM$, average token stock is $M = 400/20 = \text{CNY } 20\text{B}$ equivalent.
3. At 100% reserve backing, required reserves for that stock are 20B equivalent, not the full 400B payment flow.
4. At turnover 10, the same flow requires 40B. Liquidity buffers and precautionary balances may raise the stock further.

This arithmetic does not establish net new demand for US assets: users may substitute away from existing dollar deposits or other dollar instruments.

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Appendix: Why Exporters Keep Prices in Dollars

Gopinath's explanation (August 2026): the invoicing currency is a choice firms make for a reason.

- **Dollar-priced inputs:** exporters are also importers. When their inputs are priced in dollars, part of their cost is in dollars, so a depreciation of their own currency lowers their dollar costs only a little.
- **Global value chains:** the domestic value-added share of exports has declined. Imported inputs are a larger share of each export's cost, and suppliers and competitors also price in dollars.
- **Evidence across goods:** firms that rely more on imported inputs price more of their exports in dollars.
- **Contracts:** contracts typically fix a price and allow a range of quantities. The contracted price therefore affects how much is bought until the contract is repriced.

With fully flexible prices the invoicing currency would not matter. It matters because prices stay fixed for the length of a contract.

Source: Gita Gopinath, Conversations with Tyler, Ep. 291, recorded 27 August 2026.

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Appendix: A Depreciation versus an Import Tariff under Dollar Pricing [Background Context]

Illustration: a Thai exporter sells to a Japanese importer at USD 100, fixed in dollars for the contract. Unit cost is USD 50 of dollar-priced inputs plus THB 1,400 of local costs, at 35.0 THB per USD.

1. Initial unit cost: $50 + 1{,}400/35.0 = \text{USD } 90$. The margin is USD 10.
2. The price of a dollar rises 10%, to 38.5 THB. Unit cost falls to $50 + 1{,}400/38.5 = \text{USD } 86.36$, only 4% lower. The importer still pays USD 100: no pass-through while the price is fixed.
3. Instead, a hypothetical 25% import tariff with the price kept at USD 100: the importer pays $100 \times 1.25 = \text{USD } 125$, full pass-through at the border.
4. Absorbing the tariff would require a price of $100/1.25 = \text{USD } 80$, below unit cost.

Evidence on pass-through

Cavallo, Gopinath, Neiman, and Tang (2021) find near-complete pass-through of import tariffs to border prices in their sample, while exporters barely cut dollar prices when the dollar appreciated. Retail prices rose less because retailers absorbed part of the tariff.

Source: Cavallo, Gopinath, Neiman, and Tang, AER: Insights, 2021.

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